COMPILER DESIGN LABORATORY
Experiment No. 1 & 2

Exp. No. 1

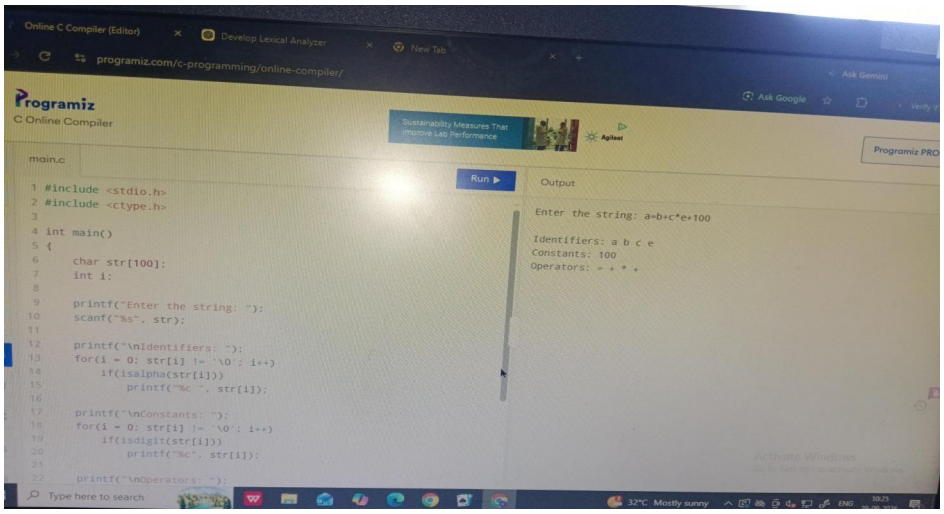
Question: Develop a lexical Analyzer to identify identifiers, constants, operators using C program.

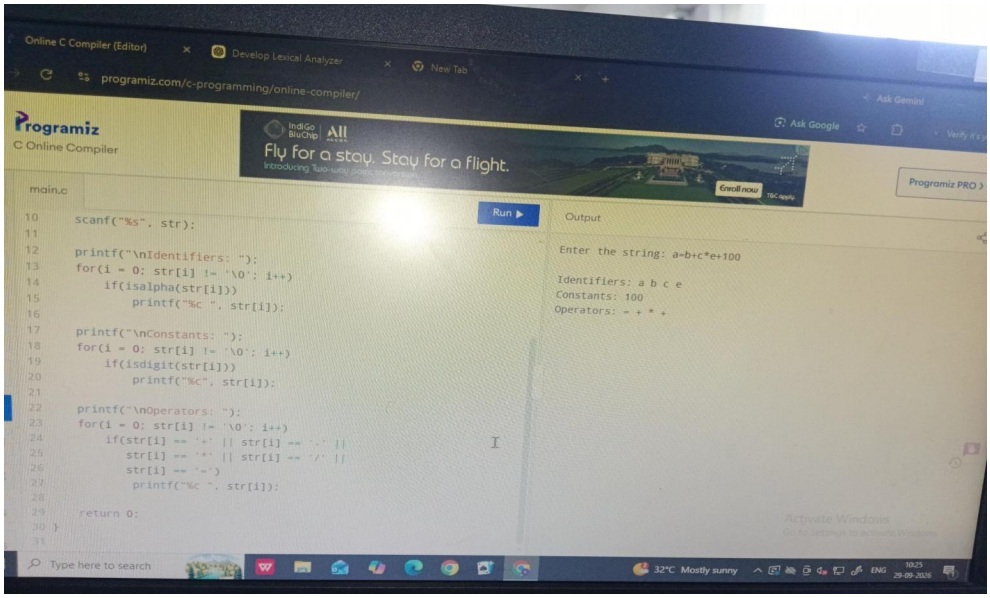
Program:

```
#include<stdio.h>
#include<ctype.h> #include<string.h> int main() { int
i,ic=0,m,cc=0,oc=0,j; char
b[30],operators[30],identifiers[30],constants[30];
printf("enter the string : "); scanf("%[^\n]s",&b);
for(i=0;i<strlen(b);i++)
{ if(isspace(b[i]))
{ continue; } else
if(isalpha(b[i]))
{ identifiers[ic]
=b[i]; ic++; } else
if(isdigit(b[i]))
{ m=(b[i]-'0');
i=i+1;
while(isdigit(b[i]))
{ m=m*10 + (b[i]-
'0'); i++; } i=i-1;
constants[cc]=m;
cc++; } else {
if(b[i]=='*')
{ operators[oc]='*';
oc++;
} else if(b[i]=='-
') {
operators[oc]='-';
oc++; } else
if(b[i]=='+') {
operators[oc]='+';
oc++; } else
if(b[i]=='=') {
operators[oc]='=';
oc++; }
} } printf(" identifiers :
"); for(j=0;j<ic;j++) {
printf("%c ",identifiers[j]);
} printf("\n constants :
"); for(j=0;j<cc;j++) {
printf("%d ",constants[j]);
} printf("\n operators :
```

```
"); for(j=0;j<oc;j++) {
printf("%c ",operators[j]);
}
```

Output:






```
enter the string : a = b + c * e + 100
 identifiers : a b c
e constants : 100
operators : = + * +
```

Exp. No. 2

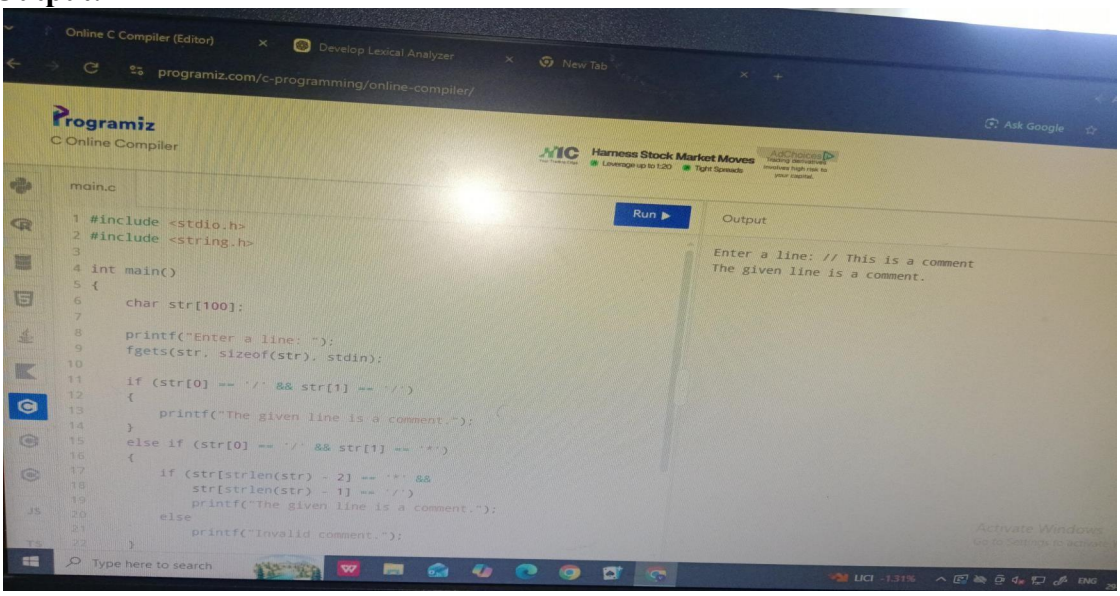
Question: Develop a lexical Analyzer to identify whether a given line is a comment or not using C.

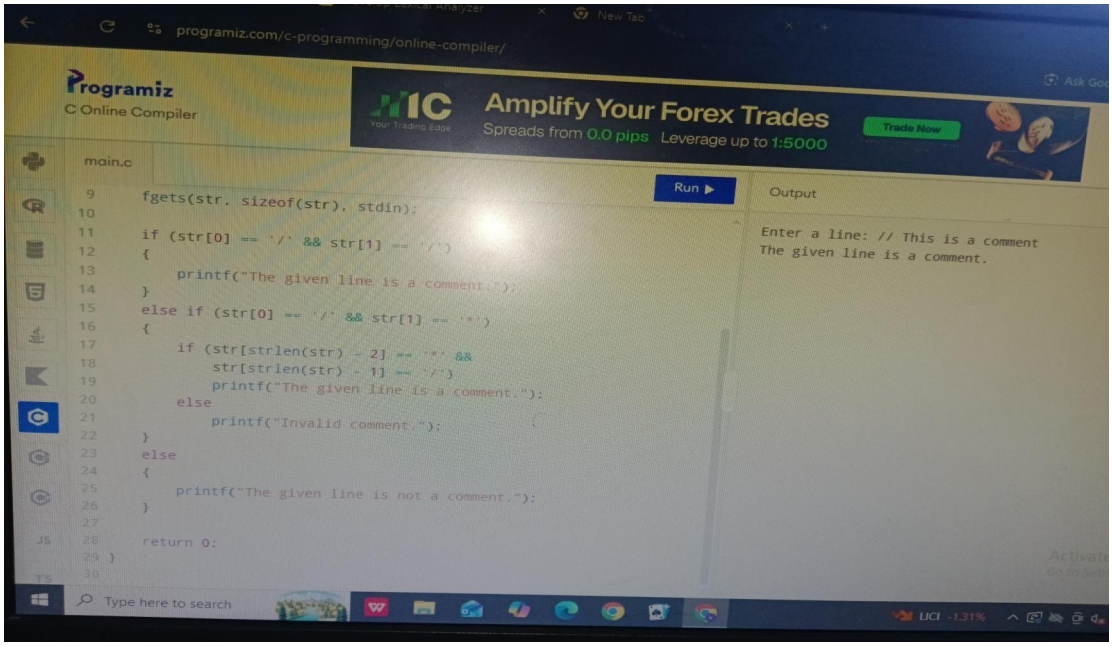
Program:

```
#include<stdio.h>
#include<conio.h> int main()
{ char com[30]; int i=2,a=0;
printf("\n Enter comment:");
gets(com); if(com[0]=='/')
{ if(com[1]=='/') printf("\n
It is a comment"); else
if(com[1]=='*')
{ for(i=2;i<=30;i++)
{ if(com[i]=='*'&&com[i+1]=='/')
{ printf("\n The given line is a
comment"); a=1; break; } else
continue; } if(a==0) printf("\n
```

```
The given line is not a
comment");
} else printf("\n The given line
is not a comment");
} else
printf("\n The given line is not a comment");
}
```

Output:





Enter comment: // This is a comment
The given line is a comment